

Clinical_feature_selection

Zhixin Mao

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```
library(readxl)
library(reshape2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(broom)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats   1.0.0      v stringr   1.5.1
## v lubridate 1.9.4      v tibble   3.2.1
## v purrr     1.0.4      v tidyr    1.3.1
## v readr     2.1.5
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggpubr)
```

```
df <- data.frame(read_excel("./data/Supplementary_table.xlsx",sheet = 1))
names(df)
```

```
## [1] "Donor.ID"          "Age"
## [3] "T.stage"           "LN.status.at.diagnosis"
## [5] "Histology"         "ER.Allred"
```

```
## [7] "ER.status" "HER2.status"
## [9] "Grade.pre.NAT" "NAT.regimen"
## [11] "Chemo.cycles" "aHER2.cycles"
## [13] "Surgery.type" "LVI"
## [15] "Tumour.dimension.surgery.1" "Tumour.dimension.surgery.2"
## [17] "Percent.cellularity" "Percent.CIS"
## [19] "Number.of.LN.removed" "Number.of.positive.LN"
## [21] "Max.LN.met.size" "RCB.score"
## [23] "RCB.category" "pCR.RD"
## [25] "PAM50" "ic10"
## [27] "DNA.sequenced" "RNA.sequenced"
## [29] "Digital.pathology"
```

```
# 161 of 168 cases had an RCB assessment
df <- df[which(df$RCB.category!="NA"),]
nrow(df)
```

```
## [1] 161
```

```
y <- df
rownames(y) <- y$Donor.ID
# define variables to use
clinVariables <- c("Age", "T.stage", "LN.status.at.diagnosis", "Histology",
                  "ER.status", "HER2.status", "Grade.pre.NAT",
                  "Chemo.cycles", "Number.of.LN.removed",
                  "RCB.category")

y <- y[, colnames(y) %in% clinVariables]

# Turn all RCB into 0, pCR into 1
y$RCB.category <- factor(y$RCB.category, levels=c("RCB-III", "RCB-II", "RCB-I", "pCR"), labels=c(0,0,0,1))

# Appropriately factor all variables
y$T.stage <- as.numeric(substr(y$T.stage,2,2))
y$LN.status.at.diagnosis <- factor(y$LN.status.at.diagnosis, levels = c("POS", "NEG"))
y$Histology <- factor(y$Histology=="IDC")
y$ER.status <- factor(y$ER.status, levels=c("POS", "NEG"))
y$HER2.status <- factor(y$HER2.status, levels=c("NEG", "POS"))
y$Grade.pre.NAT <- factor(y$Grade.pre.NAT, levels=c(2,3))
y$Chemo.cycles <- as.numeric(y$Chemo.cycles)
y$Number.of.LN.removed <- as.numeric(y$Number.of.LN.removed)
```

Before conducting feature selection and logistic regression, we examined the distribution of key clinical features across RCB outcome classes. This step reveals potential class imbalances and helps interpret model coefficients later.

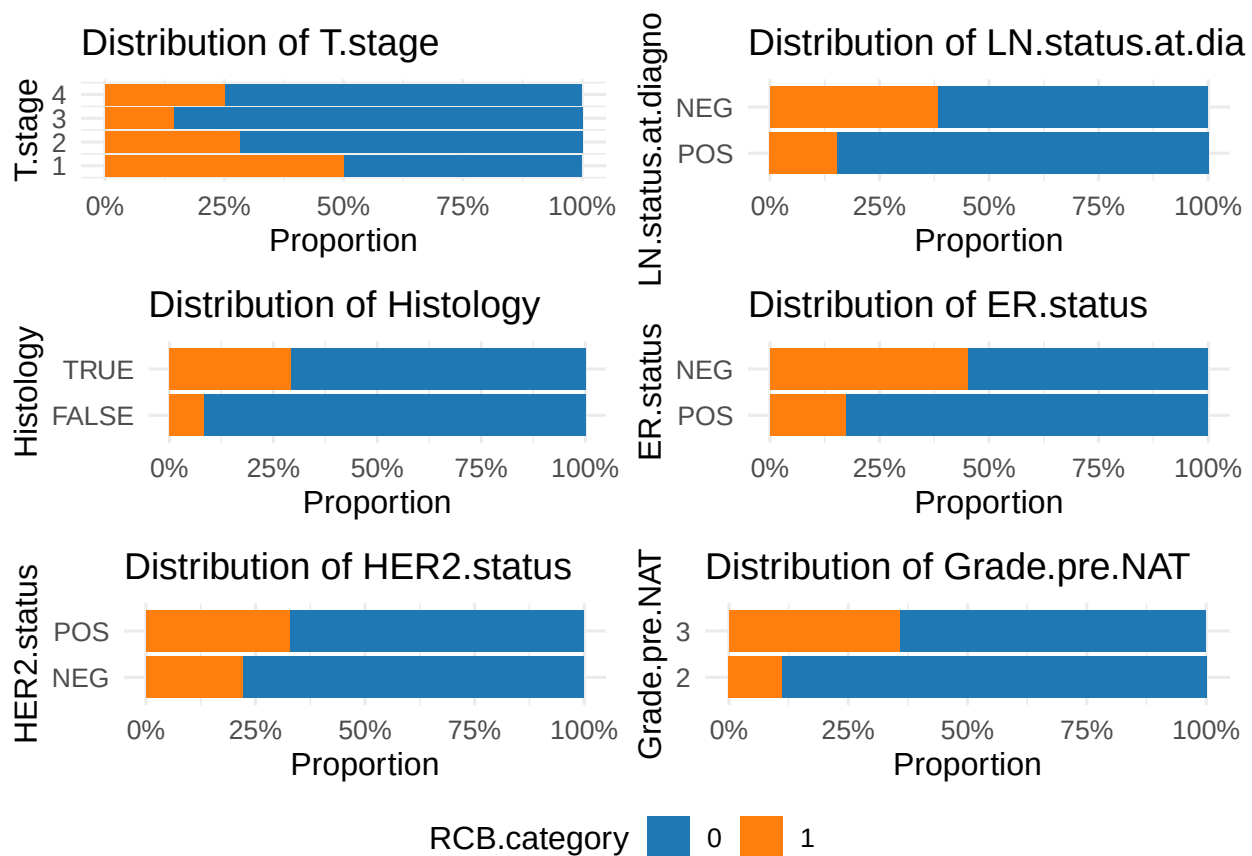
```
categorical_vars <- c("T.stage", "LN.status.at.diagnosis", "Histology",
                    "ER.status", "HER2.status", "Grade.pre.NAT")

bar_plots <- lapply(categorical_vars, function(var) {
  ggplot(y, aes_string(x = var, fill = "RCB.category")) +
    geom_bar(position = "fill") +
    scale_y_continuous(labels = scales::percent_format()) +
```

```
coord_flip() +
labs(y = "Proportion", x = var, title = paste("Distribution of", var)) +
theme_minimal(base_size = 12) +
scale_fill_manual(values = c("#1f77b4", "#ff7f0e"))
})
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
ggarrange(plotlist = bar_plots, ncol = 2, nrow = 3, common.legend = TRUE, legend = "bottom")
```

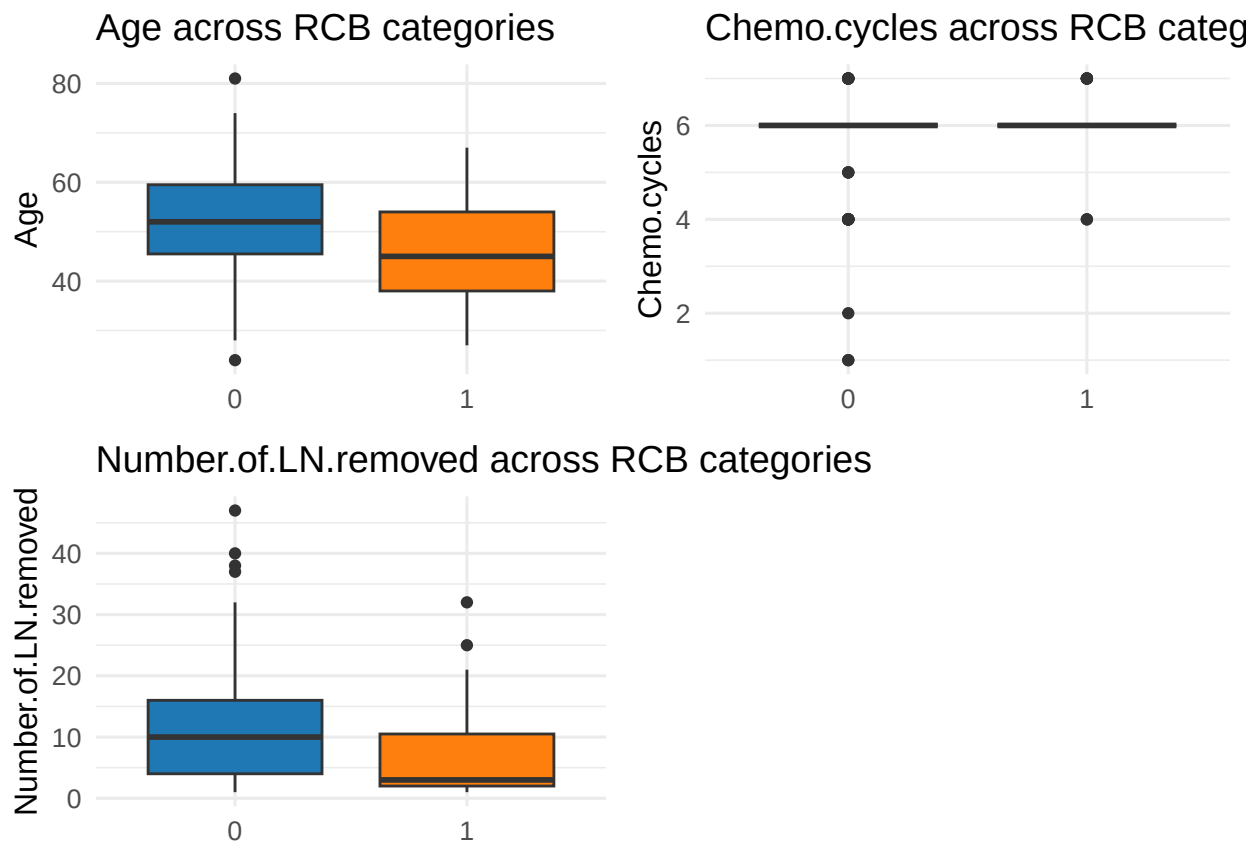


```
continuous_vars <- c("Age", "Chemo.cycles", "Number.of.LN.removed")

box_plots <- lapply(continuous_vars, function(var) {
  ggplot(y, aes_string(x = "RCB.category", y = var, fill = "RCB.category")) +
  geom_boxplot(outlier.size = 1.5) +
  labs(y = var, x = NULL, title = paste(var, "across RCB categories")) +
  theme_minimal(base_size = 12) +
  scale_fill_manual(values = c("#1f77b4", "#ff7f0e")) +
  theme(legend.position = "none")
})
```

```
})
```

```
ggarrange(plotlist = box_plots, ncol = 2, nrow = 2)
```



```
vars <- setdiff(colnames(y), "RCB.category")

# simple logistic regression
simpleLogisticRegression <- lapply(vars, function(var) {
  fml <- as.formula(paste("RCB.category ~", var))
  model <- glm(fml, data = y, family = "binomial")
  lrm <- summary(model)
  if (nrow(lrm$coefficients) < 2) return(NULL)
  pval <- lrm$coefficients[2, "Pr(>|z|)"]
  odds.ratio <- exp(lrm$coefficients[2, "Estimate"])
  conf <- suppressMessages(confint(model))
  ci.low <- exp(conf[2, 1])
  ci.hi <- exp(conf[2, 2])
  data.frame(variable = var, pval, odds.ratio, ci.low, ci.hi)
})

# combine results
regression.univariable <- do.call(rbind, simpleLogisticRegression)
regression.univariable$class <- "pCR"
regression.univariable$type <- "Simple logist."
```

```
# check results
print(regression.univariable)
```

```
##           variable           pval odds.ratio    ci.low    ci.hi class
## 1             Age 0.0055075144  0.9516227 0.9177438 0.9846738  pCR
## 2             T.stage 0.0359118623 0.5364787 0.2906491 0.9382299  pCR
## 3 LN.status.at.diagnosis 0.0013282154 3.4173486 1.6424050 7.4321932  pCR
## 4           Histology 0.0472412117 4.5360825 1.2549852 29.1435885  pCR
## 5             ER.status 0.0002894163 3.9342105 1.8872946 8.3520441  pCR
## 6           HER2.status 0.1325664269 1.7294900 0.8443116 3.5420818  pCR
## 7       Grade.pre.NAT 0.0009901803 4.4444444 1.9243582 11.6288192  pCR
## 8         Chemo.cycles 0.1080041747 1.6691200 0.9810529 3.4441356  pCR
## 9 Number.of.LN.removed 0.0040355424 0.9253199 0.8738971 0.9718752  pCR
##           type
## 1 Simple logist.
## 2 Simple logist.
## 3 Simple logist.
## 4 Simple logist.
## 5 Simple logist.
## 6 Simple logist.
## 7 Simple logist.
## 8 Simple logist.
## 9 Simple logist.
```

Interpretation

From the table, we observe that

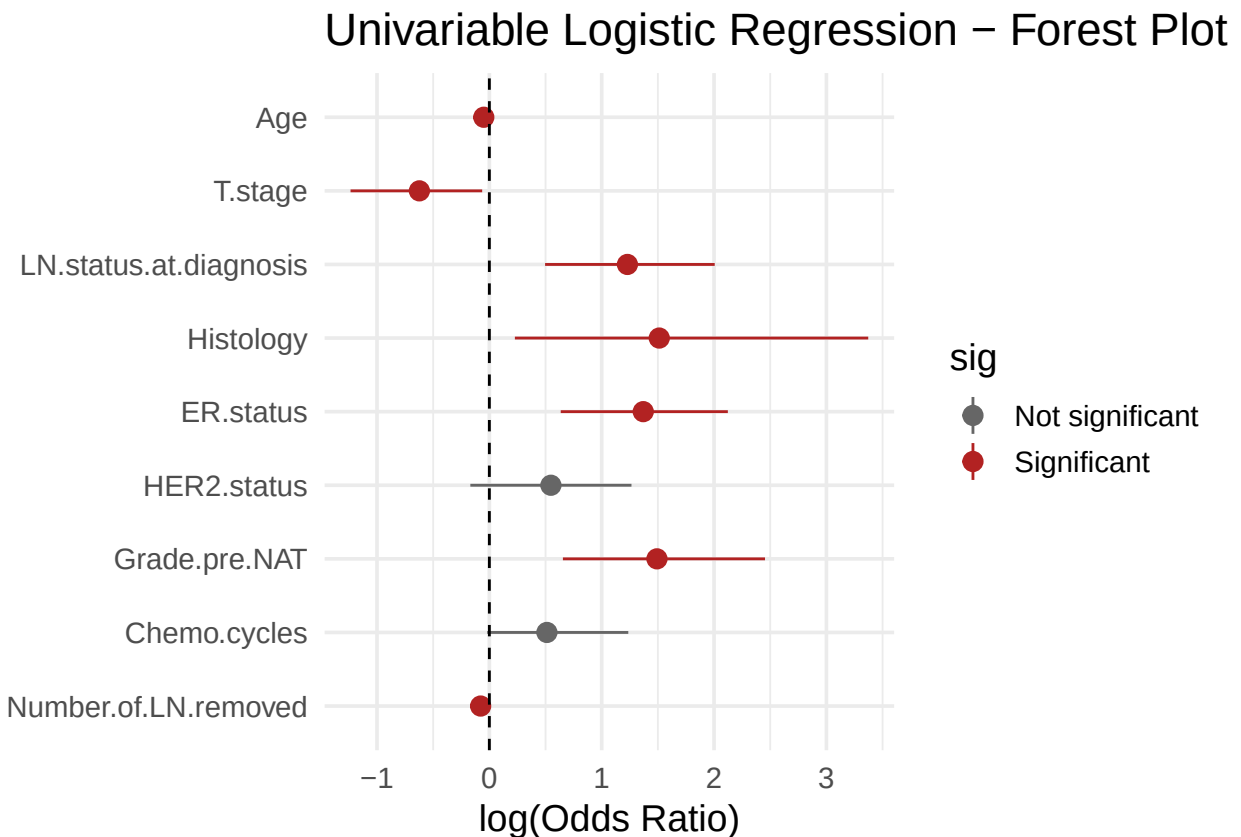
1. Older patients have lower odds of achieving pCR (−4.8% per year) — significant.
2. Larger tumor stage → significantly lower pCR odds. This is expected clinically.
3. In Ln status, there's significantly higher pCR rate in LN-negative tumors.
4. IDC histology has higher pCR odds than others. But confidence interval is large, we need to be careful.
5. ER-negative tumors are much more likely to respond (consistent with biology).
6. Better response in HER2+, but not statistically significant.
7. High-grade tumors (grade 3 vs 2) → significantly better response.
8. Slight increase in pCR odds per extra chemo cycle, but not significant.
9. For number of LN removed, slight negative association (likely a surrogate for worse disease burden).

```
# Plot forest plot for simple logistic regression result

# Prepare plot data
plot_data <- regression.univariable %>%
  mutate(
    variable = factor(variable, levels = rev(variable)), # reverse for top-down
    log_odds = log(odds.ratio),                          # log(OR) for symmetry
    sig = ifelse(pval < 0.05, "Significant", "Not significant")
  )

# Create forest plot
ggplot(plot_data, aes(x = variable, y = log_odds, ymin = log(ci.low), ymax = log(ci.hi))) +
  geom_pointrange(aes(color = sig), size = 0.6) +
  geom_hline(yintercept = 0, linetype = "dashed") +
```

```
coord_flip() +
labs(
  title = "Univariable Logistic Regression - Forest Plot",
  x = NULL,
  y = "log(Odds Ratio)"
) +
scale_color_manual(values = c("Significant" = "firebrick", "Not significant" = "grey40")) +
theme_minimal(base_size = 14)
```



```
# Multiple logistic regression
model <- glm(RCB.category ~ Age + T.stage + LN.status.at.diagnosis + Histology + ER.status +
  HER2.status + Grade.pre.NAT + Chemo.cycles + Number.of.LN.removed,
  family = "binomial", data = y)

# extract regression coefficients
summary_model <- summary(model)
conf <- suppressMessages(confint(model))[-1, , drop = FALSE] # Intercept

# variables in summary
regression_multi <- data.frame(
  variable = rownames(summary_model$coefficients)[-1],
  estimate = summary_model$coefficients[-1, "Estimate"],
  pval = summary_model$coefficients[-1, "Pr(>|z|)"],
  odds.ratio = exp(summary_model$coefficients[-1, "Estimate"]),
  ci.low = exp(conf[, 1]),
```

```

    ci.high = exp(conf[, 2])
  )

  # adjust p-value Benjamini-Hochberg
  regression_multi$padj <- p.adjust(regression_multi$pval, method = "BH")

  # mark significance
  regression_multi$sig <- ifelse(regression_multi$padj < 0.05, "Significant", "Not significant")

  # set graph order
  regression_multi$variable <- factor(regression_multi$variable, levels = rev(regression_multi$variable))

  summary_model

```

```

##
## Call:
## glm(formula = RCB.category ~ Age + T.stage + LN.status.at.diagnosis +
##      Histology + ER.status + HER2.status + Grade.pre.NAT + Chemo.cycles +
##      Number.of.LN.removed, family = "binomial", data = y)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.76036     2.84278  -0.619 0.535759
## Age             -0.05864     0.02213  -2.650 0.008042 **
## T.stage         -0.48373     0.35982  -1.344 0.178833
## LN.status.at.diagnosisNEG -0.06005     0.70225  -0.086 0.931859
## HistologyTRUE     0.72676     0.89626   0.811 0.417433
## ER.statusNEG      1.57423     0.46779   3.365 0.000765 ***
## HER2.statusPOS    0.93518     0.45667   2.048 0.040579 *
## Grade.pre.NAT3    0.98509     0.50621   1.946 0.051656 .
## Chemo.cycles      0.45515     0.36630   1.243 0.214025
## Number.of.LN.removed -0.04364     0.04460  -0.979 0.327820
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 184.82  on 160  degrees of freedom
## Residual deviance: 139.91  on 151  degrees of freedom
## AIC: 159.91
##
## Number of Fisher Scoring iterations: 6

```

```
print(regression_multi)
```

```

##              variable      estimate      pval
## Age              Age -0.05864211 0.0080416507
## T.stage          T.stage -0.48372698 0.1788327280
## LN.status.at.diagnosisNEG LN.status.at.diagnosisNEG -0.06004670 0.9318590407
## HistologyTRUE      HistologyTRUE 0.72676326 0.4174326946
## ER.statusNEG        ER.statusNEG 1.57422741 0.0007648019
## HER2.statusPOS      HER2.statusPOS 0.93517848 0.0405790178

```

```
## Grade.pre.NAT3          Grade.pre.NAT3  0.98508529 0.0516558813
## Chemo.cycles            Chemo.cycles   0.45515139 0.2140245744
## Number.of.LN.removed    Number.of.LN.removed -0.04364355 0.3278195031
##          odds.ratio    ci.low    ci.high    padj
## Age          0.9430442 0.9010649 0.9833697 0.036187428
## T.stage      0.6164815 0.2957824 1.2298111 0.321036862
## LN.status.at.diagnosisNEG 0.9417206 0.2305063 3.6956442 0.931859041
## HistologyTRUE 2.0683750 0.4114555 16.0304305 0.469611781
## ER.statusNEG  4.8270109 1.9688658 12.4736503 0.006883217
## HER2.statusPOS 2.5476681 1.0562827 6.4083864 0.116225733
## Grade.pre.NAT3 2.6780403 1.0268741 7.6576811 0.116225733
## Chemo.cycles  1.5764120 0.8263717 3.4835949 0.321036862
## Number.of.LN.removed 0.9572951 0.8720424 1.0389089 0.421482218
##          sig
## Age          Significant
## T.stage      Not significant
## LN.status.at.diagnosisNEG Not significant
## HistologyTRUE Not significant
## ER.statusNEG  Significant
## HER2.statusPOS Not significant
## Grade.pre.NAT3 Not significant
## Chemo.cycles  Not significant
## Number.of.LN.removed Not significant
```

Interpretation

The top is the summary table of multivariate logistic regression before adjusting p-value, and bottom is after adjusting p-value.

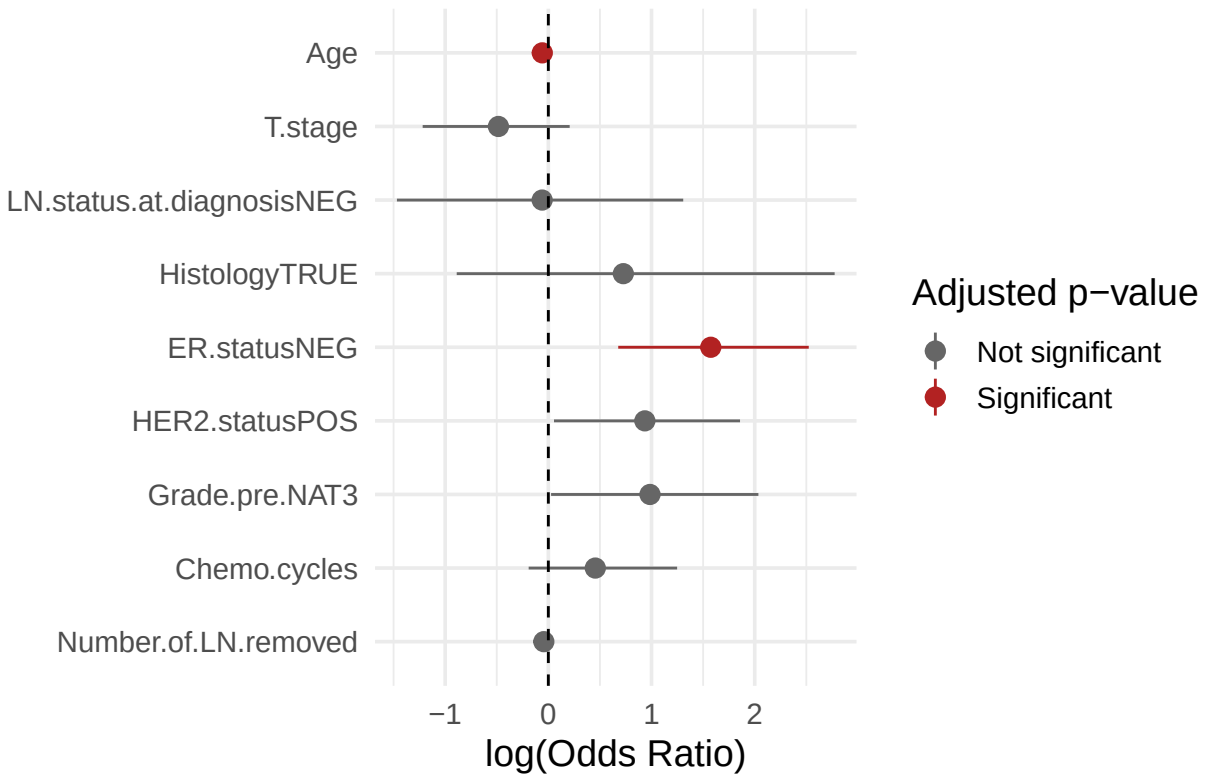
The number of statistically significant features have decreased after adjusting p value:

1. Age: Older age → less likely to achieve pCR
2. ER.statusNEG: ER-negative → much more likely to achieve pCR

```
``` r
ggplot(regression_multi, aes(x = variable, y = log(odds.ratio),
 ymin = log(ci.low), ymax = log(ci.high))) +
 geom_pointrange(aes(color = sig), size = 0.6) +
 geom_hline(yintercept = 0, linetype = "dashed") +
 coord_flip() +
 labs(
 title = "Multiple Logistic Regression - Forest Plot",
 x = NULL,
 y = "log(Odds Ratio)",
 color = "Adjusted p-value"
) +
 scale_color_manual(values = c("Significant" = "firebrick", "Not significant" = "grey40")) +
 theme_minimal(base_size = 14)
```



## Multiple Logistic Regression – Forest Plot



Next we perform lasso regression to set the coefficients of some features to 0, to help us choose features for multi-omics study.

```
library(glmnet)

Loading required package: Matrix

##
Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':
##
expand, pack, unpack

Loaded glmnet 4.1-9

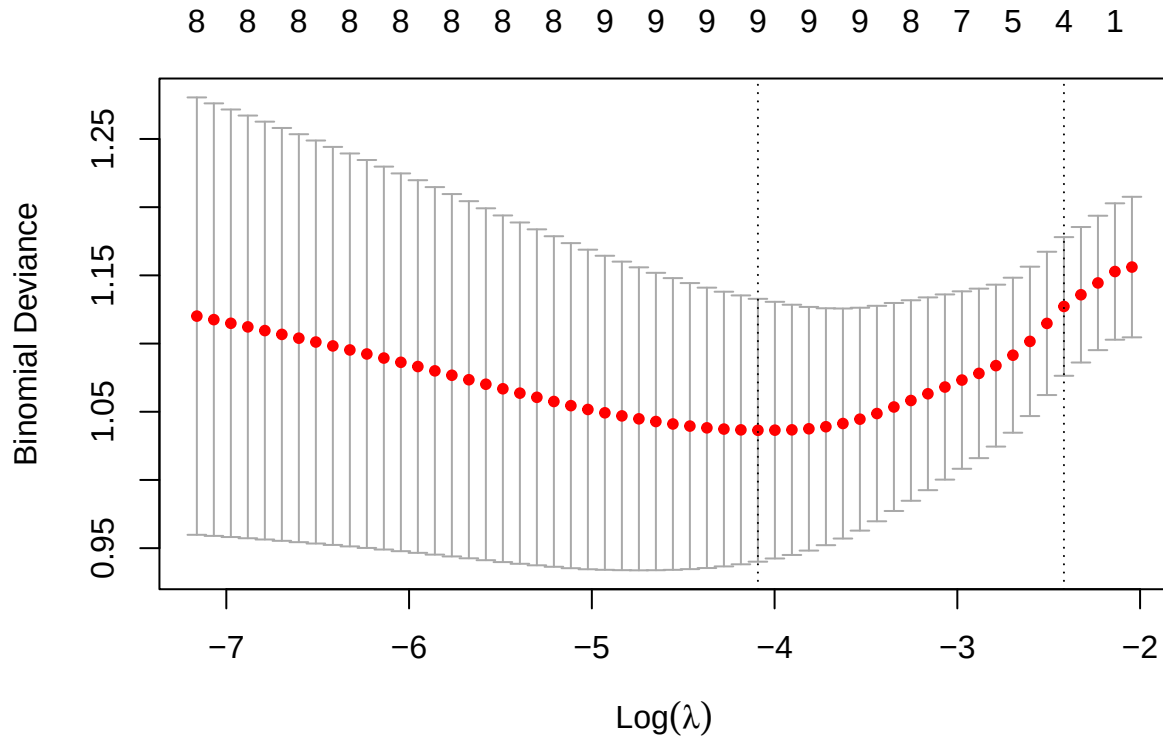
Perform Lasso Regression
X <- model.matrix(RCB.category ~ Age + T.stage + LN.status.at.diagnosis + Histology +
 ER.status + HER2.status + Grade.pre.NAT +
 Chemo.cycles + Number.of.LN.removed, data = y)[, -1]

y_bin <- as.numeric(as.character(y$RCB.category))

set.seed(123)
```

```
10-fold cross-validation to choose best lambda
cv.lasso <- cv.glmnet(X, y_bin, alpha = 1, family = "binomial")

Plot CV curve
plot(cv.lasso)
```



```
best_lambda_1se <- cv.lasso$lambda.1se
lasso_model_1se <- glmnet(X, y_bin, alpha = 1, family = "binomial", lambda = best_lambda_1se)

coef(lasso_model_1se)
```

```
10 x 1 sparse Matrix of class "dgCMatrix"
s0
(Intercept) -1.173860619
Age -0.003824679
T.stage .
LN.status.at.diagnosisNEG 0.122252982
HistologyTRUE .
ER.statusNEG 0.363980121
HER2.statusPOS .
Grade.pre.NAT3 0.225591555
Chemo.cycles .
Number.of.LN.removed .
```

According to lasso results, select the features with non-zero coefficients. The clinical features selected for multi-omics study are:

- Age
- LN.status.at.diagnosis
- ER.status
- Grade.pre.NAT3

Next, we calculate the statistics of these features.

```
Age summary
age_stats <- summary(df$Age)
cat("Age - Median:", median(df$Age, na.rm = TRUE),
 "| Min:", min(df$Age, na.rm = TRUE),
 "| Max:", max(df$Age, na.rm = TRUE), "\n")
```

```
Age - Median: 51 | Min: 24 | Max: 81
```

```
LN status proportions
ln_props <- round(prop.table(table(df$LN.status.at.diagnosis)) * 100, 2)
print(ln_props)
```

```
##
NEG POS
47.2 52.8
```

```
ER status proportions
er_props <- round(prop.table(table(df$ER.status)) * 100, 2)
print(er_props)
```

```
##
NEG POS
31.68 68.32
```

```
Grade.pre.NAT summary
grade_counts <- table(df$Grade.pre.NAT)
grade_proportion <- prop.table(grade_counts) * 100

print(grade_counts)
```

```
##
2 3
63 98
```

```
print(round(grade_proportion, 2))
```

```
##
2 3
39.13 60.87
```